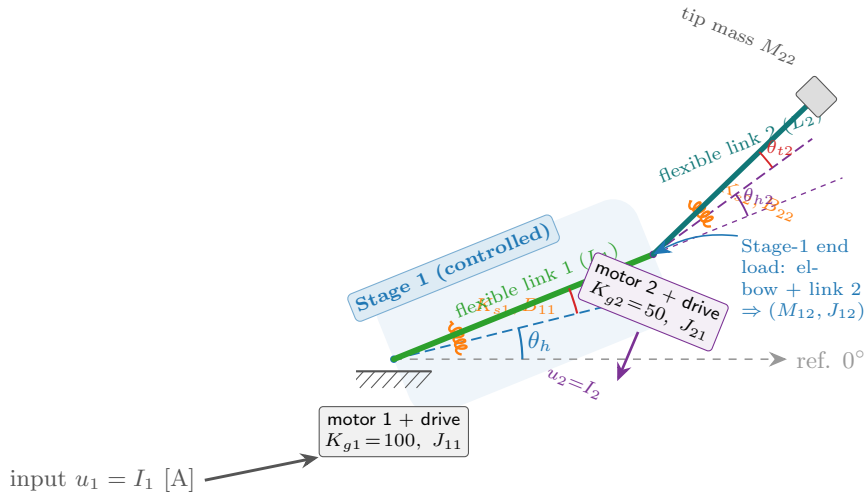




2-DOF serial flexible link — top view, horizontal plane (gravity $\perp$ page).

Inputs: motor currents $u_1 = I_1$, $u_2 = I_2$ ($|I_i| \leq 1$ A).

Sensed per joint: encoder $\rightarrow$ hub angle (θ_h, θ_{h2}) , strain gauge $\rightarrow$ deflection (θ_t, θ_{t2}) .

Stage-1 model (compare_methods.py) — joint 1 only:

$$\mathbf{x} = [\theta_h \quad \theta_t \quad \dot{\theta}_h \quad \dot{\theta}_t]^\top, \quad \text{input } u_1$$

output $y = \theta_h + \theta_t$ — absolute link-1 tip angle

K_{s1}, B_{11} root flex; J_{11} hub; M_{12}, J_{12} end load

MPC adds the state limit $|\theta_t| \leq \theta_{\max}$